REPRODUCTION OF IMAGE-INPAINTING MODEL BASED ON FRACTIONAL-ORDER NON-LINEAR DIFFUSION

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Introduction

Importance:

- *Image Inpainting Tasks:*
  - Restores missing or corrupted image regions.
  - Applications: Photo restoration, object removal, and image compression.
- *Traditional Techniques*:
  - <u>Exemplar-based</u>: Synthesizes missing areas using similar patches.
  - <u>Diffusion-based</u>: Propagates information along intensity gradients.

Challenges:

- Staircase effects in second-order diffusion.
- Loss of edge connectivity and blurring.

Why Fractional-Order Nonlinear Diffusion?

- **Innovations:**
  - Introduces **fractional-order PDEs** for better balance:
    - Retains edges while minimizing artifacts like staircase effects.
    - Reduces speckle noise effectively.
- **Enhanced Features:**
  - Incorporates **Difference Curvature (DC):**
    - Differentiates edges from smooth regions.
    - Guides accurate inpainting processes.
  - Adaptive **Conductance Coefficients(ψ_{Ω}):**
    - Controls the strength of diffusion.
    - High in smooth regions to encourage diffusion.
    - Low near edges to prevent smoothing.

Original Paper: Proposed Methodology

Problem Statement:

- Given an image, $u_0 \in L^2(\Omega),\ \ \Omega \subset \mathbb{R}^2$. The Ω which is the inpainting region has a small boundary of $\delta\Omega$.
- Our problem is reconstruct $u(x, y)$ from $u_0(x, y)$. The degraded image is:

$$u_0(x, y) = h(x, y) * u(x, y) \ + \ n(x, y)$$

where, $h(x, y) = \frac{1}{2\pi\sigma^2}.\exp\left(-\frac{x^2+y^2}{2\sigma^2}\right)$ is a bounded linear shift-invariant operator , and $n(x, y)$ is gaussian noise of zero mean and finite variance of σ^2.

Original Paper: Proposed Methodology

The inpainting process is guided by the following partial differential equation:

$$\frac{\partial u}{\partial t} = -D^{\alpha}(\psi_{\Omega} D^{\alpha} u) + \lambda_{\Omega} h(h * u \ - u_0)$$

Where:

- The term $-\boldsymbol{D^{\alpha}(\psi_{\Omega} D^{\alpha} u)}$:
    - $D^{\alpha} u$ is the Fractional-order derivative operator of the 2D-signal which uses DFT.
    - Applies fractional-order diffusion, driven by the conductance coefficient ψ_{Ω}, which depends on DC.
- The term $\boldsymbol{\lambda_{\Omega} h(h * u \ - u_0)}$:
    - Restores the non-degraded regions based on the degradation model $h(x,y) * u(x,y) \ + \ n(x,y)$
- The Conductance coefficient $\psi_{\Omega} = f(DC)$ is defined as:

Where,

$k = 1.4826 \ * MAD(|D^{\alpha} u|) = 1.4826 \ * median(|D^{\alpha} u| - \text{median}(|D^{\alpha} u|))$

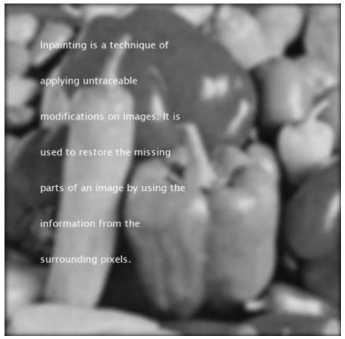
Algorithm WorkFlow:

- Initialization:
  - $u^0 = u, \Delta t = 0.01, \lambda^0 = 0, DC^0 = 0, \alpha = \begin{cases} \alpha_1, & (x,y) \in \Omega \\ \alpha_2, & (x,y) \notin \Omega \end{cases}, \hat{u}^0 = DFT\{u^0\}$
- Iterations:
  - Use 2D DFT to compute Fractional order derivatives $D^\alpha u(x,y)$ along x, y directions
  - Compute $u_{\eta\eta}$ & $u_{\xi\xi}$, the second-order derivatives in tangential and perpendicular directions. Calculate $DC = ||u_{\eta\eta}| - |u_{\xi\xi}||$.
  - Adaptively compute $\psi_\Omega = f(DC)$
  - Evolve the image $u(x,y)$ over iterations m using

$$u^{m+1} = u^m + \Delta t.(-D^\alpha(\psi_\Omega D^\alpha u) + \lambda_\Omega h(h * u - u_0))$$

Results

Original Image



Degraded Image

Proposed Model Output

TV output

Bilateral Output

Model	**PSNR**	**MSSIM**	**FoM**
Proposed	34.7995	0.9765	0.9529
TV	25.2140	0.8989	0.5661
Bilateral Telea	24.6290	0.7617	0.3243

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Challenges

- **Mask Generation:**
  - Original paper lacked details on generating masks for degraded regions.
  - Solution: Implemented **edge-driven mask generation**:
    - Uses edge detection to identify prominent areas.
    - Applies selective occlusions to create masks.
  - Considered an **extension** to the proposed work.
- **Stopping Criterion:**
  - Original criterion based on PSNR lacked clarity for convergence.
  - Solution:
    - Refined criterion using **ΔPSNR**:
      - $\Delta PSNR = PSNR(u^{m+1}, u_0) - PSNR(u^m, u_0)$
      - Threshold: 1×10^{-2}
    - Introduced iteration counter for **stability**.
- **Kernel Size:**
  - No kernel size specification in the original paper.
  - Solution:
    - Conducted experiments with various kernel sizes.
    - Optimized for computational efficiency and result quality.

Extension Work

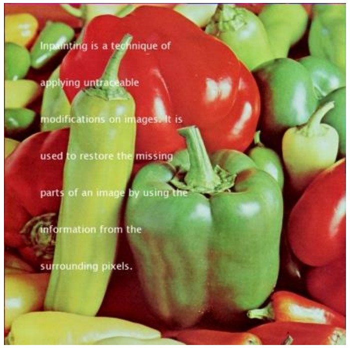
- **Handling RGB Images:**
  - **Channel Separation Approach :**
    - **Method**:
      - Split RGB images into individual grayscale channels (Red, Green, Blue).
      - Apply the inpainting algorithm independently to each channel, then recombine them.
    - **Advantages**:
      - Simple to implement, leveraging the original 2D inpainting method.
      - Computational complexity remains similar to the original model.
    - **Disadvantages**:
      - Potential color artifacts due to inconsistencies between channels.
      - Does not fully exploit the interdependencies of RGB channels.




Extension Work

- **Handling RGB Images:**
  - **Direct 3D Approach:**
    - **Method**:
      - Treat the RGB image as a 3D signal by adding a third dimension for color channels.
      - Modify conductance functions and equations to account for interactions between channels.
    - **Advantages**:
      - Maintains color coherence and structural integrity across RGB channels.
      - Better visual quality, especially for complex color details.
    - **Disadvantages**:
      - Higher computational complexity.
      - Requires significant modifications to the original algorithm.

Original Image

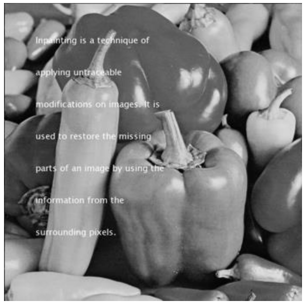


Degraded Image.

Restored Image

Extension Work

- **Edge Driven Mask Generation:**
  - **Motivation**
    - **Challenge**: Masks are often unavailable in real-world scenarios.
    - **Solution**: Developed an automatic mask generation approach based on edge detection to identify inpainting regions.
  - **Methodology:**
    - **Preprocessing**: Convert to grayscale for consistent edge detection.
    - **Edge Detection**: Use **Sobel operator** to find intensity changes.
    - **Morphological Dilation**: Expand edges with a disk-shaped structuring element.
    - **Mask Creation**: Generate a binary mask for inpainting regions.
    - **Visualization**: Save and visualize the mask.



Degraded Image

Generated Mask

Conclusion

- Reproduced and extended the fractional-order nonlinear diffusion-based inpainting model.
- **Key Contributions:**
  - Extended model to handle RGB images using channel separation and 3D signal processing.
  - Developed edge-driven mask generation for automatic identification of degraded regions.
- **Performance Highlights:**
  - Achieved superior PSNR, MSSIM, and FoM compared to traditional methods (TV model, Bilateral Telea).
  - Demonstrated robust handling of noisy and degraded images with structural and color coherence.
- **Future Directions:**
  - Optimize computational efficiency and explore adaptive parameter tuning.
  - Extend applications to hyperspectral/satellite images and integrate machine learning for enhanced adaptability.